

Verification of whether the final matrix is in rref.

1. Row 1 and 2 are non-zero rows. Row 3 is a zero row. $\therefore$ The condition that all the non-zero rows must be above the zero row is True.

2. Row 1 Leading entry = 1 in Column 1
Row 2 Leading entry = 1 in Column 2

$\therefore$ The condition that for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it is True.

3. Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0

$\therefore$ The condition that for each column having a leading entry, all positions below it must contain 0 is True.

4. The condition that each leading entry must be 1 is True.

5. Row 1 Column 2 = 0
 $\therefore$ The condition that for each column having a leading entry, all positions above it must contain 0 is True.

So, the final matrix is in rref $\square$

$\therefore$ The given matrix cannot be row reduced to ~~row~~ I.

By the Invertible Matrix Theorem and $\nexists A^{-1}$, A is not an invertible matrix. $\square$

$$\begin{array}{l}
 7. \left[\begin{array}{cccc} -1 & -3 & 0 & 1 \\ 3 & 3 & 8 & -3 \\ -2 & -6 & 3 & 2 \\ 0 & -1 & 2 & 1 \end{array} \right] \xrightarrow[\times(-1)]{\text{Row 1} = \text{Row 1}} \left[\begin{array}{cccc} 1 & 3 & 0 & -1 \\ 3 & 3 & 8 & -3 \\ -2 & -6 & 3 & 2 \\ 0 & -1 & 2 & 1 \end{array} \right] \xrightarrow[\text{-3} \times \text{Row 1}]{\text{Row 2} = \text{Row 2}} \\
 \left[\begin{array}{cccc} 1 & 3 & 0 & -1 \\ 0 & -4 & 8 & 0 \\ -2 & -6 & 3 & 2 \\ 0 & -1 & 2 & 1 \end{array} \right] \xrightarrow[\text{+2} \times \text{Row 1}]{\text{Row 3} = \text{Row 3}} \left[\begin{array}{cccc} 1 & 3 & 0 & -1 \\ 0 & -4 & 8 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & -1 & 2 & 1 \end{array} \right] \xrightarrow[\times 1/(-4)]{\text{Row 2} = \text{Row 2}} \left[\begin{array}{cccc} 1 & 3 & 0 & -1 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & -1 & 2 & 1 \end{array} \right] \\
 \left[\begin{array}{cccc} 1 & 3 & 0 & -1 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow[\text{+Row 2}]{\text{Row 4} = \text{Row 4}} \left[\begin{array}{cccc} 1 & 3 & 0 & -1 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]
 \end{array}$$